

MOLDE ARO, Norway

WMO: 012170

Lat: 62.745N

Lon: 7.263E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-6.7	-13.3	1.2	-5.0	-11.5	1.4	-4.4	14.9	3.8	12.7	4.5	4.3	30	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	24.0	17.1	22.0	16.1	20.2	15.3	18.0	22.1	16.8	20.6	15.9	19.4	3.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.4	20.4	15.0	10.7	18.9	14.1	10.1	17.7	50.7	22.3	47.2	20.7	44.4	19.5	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.8	10.7	9.1	-10.0	28.1	1.8	2.1	-11.3	29.7	-12.3	30.9	-13.3	32.1	-14.6	33.7		
			-10.5	19.9	1.8	1.6	-11.8	21.1	-12.8	22.1	-13.8	23.0	-15.1	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	0.6	1.8	2.9	5.7	9.8	12.9	15.5	14.8	11.8	7.4	3.4	2.3	
	DBStd	6.08	3.43	3.44	3.08	3.15	3.84	2.89	3.03	2.53	2.65	3.07	3.70	3.60	
	HDD10.0	1473	291	229	219	134	52	6	1	1	13	92	198	238	
	HDD18.3	3988	549	463	477	379	266	164	98	113	196	340	447	496	
	CDD10.0	542	0	0	0	6	45	94	172	149	67	9	1	0	
	CDD18.3	17	0	0	0	0	1	2	12	3	0	0	0	0	
	CDH23.3	85	0	0	0	0	4	12	59	10	0	0	0	0	
	CDH26.7	11	0	0	0	0	0	1	9	1	0	0	0	0	
Wind	WSAvg	4.0	4.4	4.2	4.5	3.8	3.5	3.6	3.2	3.2	3.9	3.9	4.3	5.0	
Precipitation	PrecAvg	1664	158	154	161	85	85	107	100	126	170	171	161	185	
	PrecMax	2136	364	365	463	225	167	251	193	303	448	315	386	389	
	PrecMin	1094	0	0	34	7	25	25	26	17	43	20	2	26	
	PrecStd	256	106	92	108	57	39	58	46	65	105	73	112	83	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	10.7	11.9	19.8	23.8	25.0	28.0	24.5	20.6	17.4	13.0	11.2	
		MCWB	5.6	6.2	6.3	11.6	15.5	16.9	19.1	17.3	14.6	11.9	9.3	7.1	
	2%	DB	7.5	9.0	10.0	15.8	21.1	22.1	24.9	22.1	18.6	15.2	11.1	9.6	
		MCWB	5.2	5.4	6.0	9.8	14.4	15.6	18.1	16.0	13.6	10.7	8.2	6.5	
	5%	DB	6.3	7.7	8.7	13.0	18.2	19.7	22.6	20.5	17.2	13.0	9.7	8.4	
		MCWB	4.2	4.7	5.8	8.6	12.9	14.2	17.3	15.8	13.3	9.7	7.4	5.7	
	10%	DB	5.1	6.5	7.6	10.9	15.9	17.9	20.3	19.0	15.9	11.6	8.2	7.2	
		MCWB	3.0	3.9	4.9	7.1	11.5	13.2	16.0	15.1	12.7	9.0	6.1	4.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	7.5	8.0	12.1	16.1	17.2	20.4	18.1	16.1	12.7	10.2	8.5
			MCDB	8.4	9.1	9.6	18.2	22.1	22.7	25.9	22.0	19.0	15.5	12.0	9.8
2%		WB	5.7	6.2	7.1	10.4	14.7	15.8	18.9	17.1	14.8	11.4	9.1	7.3	
		MCDB	6.9	7.9	8.9	15.1	19.8	20.9	23.8	20.6	17.3	14.3	10.8	8.9	
5%		WB	4.4	5.2	6.0	9.1	13.3	14.7	17.6	16.2	13.9	10.2	7.6	6.2	
		MCDB	5.9	7.3	8.5	12.4	17.6	19.0	21.6	19.3	16.2	12.7	9.2	7.8	
10%		WB	3.4	4.1	5.0	7.6	12.0	13.6	16.5	15.4	13.1	9.1	6.4	5.2	
		MCDB	4.7	6.2	7.1	10.1	15.6	17.1	20.0	18.4	15.4	11.2	7.9	7.2	
Mean Daily Temperature Range		5% DB	MDBR	4.7	4.9	6.2	7.5	7.9	6.9	6.8	6.9	6.3	5.9	4.8	4.7
			MCDBR	5.6	6.0	7.1	11.4	11.7	10.5	10.2	9.4	7.9	7.4	6.2	6.2
	MCWBR		4.2	4.1	4.7	6.6	6.3	5.6	5.3	5.0	4.8	4.9	4.7	4.5	
	5% WB	MCDBR	5.2	5.4	6.4	10.2	10.8	9.3	9.5	8.3	7.0	6.5	5.6	6.0	
		MCWBR	4.1	4.1	4.6	6.1	6.0	5.3	5.2	4.8	4.7	4.5	4.7	4.7	
Clear-Sky Solar Irradiance	taub	0.241	0.296	0.324	0.347	0.347	0.350	0.360	0.351	0.331	0.307	0.249	0.358		
	taud	2.082	2.265	2.317	2.355	2.414	2.468	2.479	2.502	2.539	2.452	2.147	2.597		
	Ebn at Noon	406	652	778	838	870	874	853	830	780	649	393	235		
	Edh at Noon	32	60	83	98	100	97	94	85	68	50	29	20		
All-Sky Solar Radiation	RadAvg	0.14	0.61	1.70	3.32	4.56	4.88	4.57	3.50	2.05	0.93	0.24	0.08		
	RadStd	0.03	0.13	0.27	0.36	0.28	0.49	0.39	0.34	0.28	0.14	0.06	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)		+0.50	+0.68	N/A	N/A	N/A	N/A	-150	-219	+63	N/A

Nomenclature: See separate page